

CS Honors Program Review Committee

Department of Computer Science
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Dear Members of the Committee,

I am a rising senior pursuing a B.S. in Computer Science on the Computational Biology track, and I am writing today to propose an honors thesis titled “Privacy-Preserving Phenotype Matching for Rare Disease Cohorts.” The goal is to develop and rigorously evaluate a coordinated discovery tool that allows peer institutions to discover “patients like mine” using standardized phenotypes for further gene research and clinical treatment without sharing raw records, genomes, or other protected health information (PHI). Professor Stephen Montgomery has agreed to oversee my work, and this thesis aligns closely with his lab’s focus on human genetics and rare-disease phenotypes.

Motivation

Diagnosing rare diseases and identifying the genes involved in their etiology hinges on identifying phenotypically similar patients across institutions. Today, that typically requires the centralization of sensitive patient information or the exchange of information-rich patient profiles. Current infrastructures present considerable privacy, regulatory, and operational barriers to effective treatment. Implementing a privacy-based foundational matching layer that provides useful matches with minimal PHI disclosure promises to expedite cohort assembly and downstream interpretation.

Related Work and Distinction

Recent systems (e.g., SHEPHERD, *npj Digital Medicine*, 2025) have demonstrated that phenotype-guided embeddings can power gene ranking and “patients like mine” identification in centralized settings. My work extends this development with the integration of formal privacy measures by computing similarity over GA4GH Phenopackets under strict statistical cryptographic and privacy safeguards.

Innovations

1. Two step “reveal ladder”
 - a. Step 1: private, thresholded similarity test and differentially private (DP) aggregate counts
 - b. Step 2: minimal, standards-compliant data slices after step 1 checks pass

2. Risk-aware phenotype encoding
 - a. Down-weight potentially re-identifying Human Phenotype Ontology (HPO) terms while preserving key information for matching
3. Cryptographic protocol test
 - a. Implement and compare Private Set Intersection (PSI) and secure sketch / locality-sensitive hashing approximate nearest neighbor (LSH-ANN) approaches for private k-NN over HPO profiles
 - b. Perform explicit leakage audits under honest-but-curious and collusion threat models
4. Standards bridge
 - a. Adapters for GA4GH Phenopackets I/O
 - b. Compatibility layers for Matchmaker Exchange (MME) and Beacon v2 (DP-guarded counts → contact)
5. Failure-mode guarantees
 - a. k-anonymity guardrails (no results if <k matches) and DP noise on outputs to mitigate re-identification of patient cohorts

Specific Targets

1. Baseline and Metrics
 - a. Expand existing open, non-private baselines, including Resnik/IC HPO similarity, simple HPO embeddings, and establish retrieval metrics (Recall@k, nDCG, calibration)
2. Target 2: Privacy-Preserving Retrieval
 - a. Implement PSI-based and secure-sketch/ANN pipelines
 - b. Quantify the privacy-utility frontier (retrieval quality vs. ϵ , k, term-prevalence filters, sparsity)
3. Target 3: Integration and Demo
 - a. Ship a containerized service with Beacon-style DP aggregates and an MME-compatible hand-off
 - b. Present existing logs, audit trails, and clear operator controls

Methods

- Representation: GA4GH Phenopackets with HPO term vectors (information-content weights) and optional compact genomic hints (candidate gene set IDs, variant class tags)
- Similarity and Learning: deterministic IC-weighted scores and optional learned HPO embeddings; calibration for top-k retrieval
- Privacy Mechanisms
 - PSI to compute overlap statistics without revealing non-intersecting terms
 - Secure sketches / LSH-ANN for approximate nearest neighbors
 - Differential privacy on counts/ranks
 - k-anonymity threshold
 - Rare-term suppression
- Threat Model and Evaluation

- Honest-but-curious sites with possible collusion
- Audits for attribute and cohort membership identification
- Report Recall@k/nDCG and compare to previously established open baseline
- Replicate SHEPHERD-style retrieval metrics under privacy budgets

Feasibility and Resources

To build a prototype model, I will utilize a combination of synthetic and public Phenopackets and plan to complete comprehensive testing without PHI, integrating de-identified institutional data if granted access. I will develop and evaluate the system along three data tracks in parallel:

1. Foundational development with Synthetic Phenopackets (immediate)
 - a. Create a script that generates synthetic GA4GH Phenopackets using HPO term prevalence with optional compact genomic hints (e.g., candidate gene sets, variant class tags)
2. Public de-identified EHR
 - a. Use MIMIC-IV (ICU EHR) for method components that benefit from real clinical structure while remaining outside PHI
 - b. Map curated findings to HPO to evaluate the full pipeline
3. Stanford de-identified EHR
 - a. Under Professor Montgomery's sponsorship, I plan to request access to STARR-OMOP-deid (BigQuery).
 - b. If granted access, I will use Cohort Discovery/Chart Review to scope phenotypic cohorts and export standards-encoded features (HPO/Phenopackets) into the private-matching pipeline.

Timeline

- Fall Quarter
 - Create project repo, milestone projections, identify key rare disease phenotypes
 - Apply for data: STARR-OMOP-deid, MIMIC-IV
 - Stand up Synthetic Phenopacket generator
 - Implement PSI-based private matching pipeline and generate first privacy-utility curves
 - **Deliverables:** Phenopackets corpus, PSI prototype, baseline metrics
- Winter Quarter
 - Implement secure-sketch / LSH-ANN private retrieval and add calibration for top-k
 - Compare PSI vs. LSH performance under sparsity, rare-term suppression, DP noise; plot privacy-utility frontier
 - Add MME adapter, k-anonymity guardrails and audit logging
 - Containerize the service and document run protocol
 - If approved, STARR-OMOP-deid cohort extraction to phenotype features only
 - **Deliverables:** preliminary report for feedback; draft finalized list of validation figures

- Spring Quarter
 - Real-data evaluation: compare private vs. open baselines on retrieval
 - Leakage audit (membership/attribute inference simulations); finalize DP ϵ/k settings and rare-term policy
 - Ablations: phenotype sparsity, mis-negations, cross-site heterogeneity
 - Prepare results and methods report
 - **Deliverables:** real-data evaluation report, leakage audit, final thesis, open-source artifact and demo

Preparation and Fit

My Stanford CS coursework has equipped me with the technical expertise to undertake this project with an academically rigorous approach and a privacy-first mindset. Additionally, my research experience at the Biobehavioral Pediatric Pain Lab and Cremer Lab at the Stanford School of Medicine have provided me with firsthand insights into how integral comprehensive, privacy-preserving data sets are to enabling healthcare providers to diagnose and treat diseases effectively. As I look ahead to a future career in medicine, it would be a privilege to have the opportunity to work to improve healthcare access and quality.

I am incredibly excited by the prospect of pursuing this work under the guidance of Professor Montgomery as the culmination of my Stanford undergraduate career. Thank you for your time and consideration.

Sincerely,

Patrick Walsh